MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, LOW POWER, NAND GATES, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, low power positive NAND logic gating microcircuits. Three product assurance classes and a choice of case outline/lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The complete part number shall be as shown in the following example:

M38510	/020	01	B	A	C
Military designator	Detail specification	Device type (1.2.1)	Device class (1.2.2)	Case outline (1.2.3)	Lead finish (3.3)

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Single, 8-input positive NAND gate
02	Dual, 4-input positive NAND gate
03	Triple, 3-input positive NAND gate
04	Quadruple, 2-input positive NAND gate
05	Hex, 1-input inverter gate
06	Quadruple, 2-input positive NAND gate (open collector output)

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline, MIL-M-38510, Appendix C
A	F-1 (14-pin, 1/4" x 1/4" flat pack)
B	F-3 (14-pin, 1/8" x 1/4" flat pack)
C	D-1 (14-pin, 1/4" x 3/4" dual-in-line pack)
D	F-2 (14-pin, 1/4" x 3/8" flat pack)

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1.3 Absolute maximum ratings:

Supply voltage range	0 Vdc to 8.0 Vdc
Input voltage range	0 Vdc to 6.0 Vdc
Storage temperature range	-65° C to 150° C
Maximum power dissipation, P_D 1/	4 mWdc per gate
Lead temperature (soldering 10 seconds)	300° C
Thermal resistance, junction to case, θ_{JC}	0.09° C/mW for flat pack 0.08° C/mW for dual-in-line pack
Junction temperature	T_J = 175° C

1.4 Recommended operating conditions:

Supply voltage	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage	2.0 Vdc
Maximum low level input voltage	0.7 Vdc
Normalized fanout (each output) 2/	10 maximum
Ambient operating temperature range	-55° C to 125° C

2. APPLICABLE DOCUMENT

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuit. The schematic circuit shall be as specified on figure 3.

3.2.4 Case outlines. The case outlines shall be in accordance with 1.2.3.

3.3 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510.

1/ Must withstand the added P_D due to short circuit conditions (e.g., I_{OS}) at one output for 5 seconds duration.

2/ Device will fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I and apply over the full recommended ambient operating temperature range, unless otherwise specified.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions	Device type	Limits		
				Min	Max	Units
High-level output voltage (except open collector output circuits)	V_{OH}	V_{CC} = 4.5V, V_{IN} = 0.7V I_{OH} = -100 µA 1/	01, 02, 03, 04, 05	2.4	---	Volts
Low-level output voltage	V_{OL}	V_{CC} = 4.5V, I_{OL} = 2 mA V_{IN} = 2.0V for all inputs of gate under test 1/	All		0.3	Volts
Maximum collector cut-off current (open collector circuits only)	I_{CEX}	V_{CC} = 4.5V, V_{IN} = 0.6V V_{OH} = 5.5V	06		50	µA
High-level input current	I_{IH_1}	V_{CC} = 5.5V, V_{IN} = 2.4V 2/	All		10	µA
High-level input current	I_{IH_2}	V_{CC} = 5.5V, V_{IN} = 5.5V 2/	All		100	µA
Low-level input current	I_{IL}	V_{CC} = 5.5V, V_{IN} = 0.3V 1/	All	-60	-180	µA
Short-circuit output current (except open collector circuits)	I_{OS}	V_{CC} = 5.5V 2/ 3/	01, 02, 03, 04, 05	-3	-15	mA
High-level supply current per gate	I_{CCH}	V_{CC} = 5.5V 2/ V_{IN} = 0V	02, 03, 04, 05, 06		200	µA
			01		330	µA
Low-level supply current per gate	I_{CCL}	V_{CC} = 5.5V 1/ V_{IN} = 5.5V	All		510	µA
Propagation delay time, high-to-low-level	t_{PHL}	C_L = 50 pF, R_L = 4KΩ	01	3	140	ns
			02, 03, 04, 05, 06	3	90	ns
Propagation delay time, low-to-high-level	t_{PLH}	C_L = 50 pF, R_L = 4KΩ	01, 02, 03, 04, 05	3	90	ns
			06	3	140	ns

1/ All unspecified inputs at 5.5 volts.
2/ All unspecified inputs grounded.
3/ Not more than one output should be shorted at a time.

3.5 Rebonding. Rebonding shall be in accordance with MIL-M-38510.

3.6 Electrical test requirements. Electrical test requirements shall be as specified in table III for the applicable device type and device class. The subgroups of table III which constitute the minimum electrical test requirements for screening, qualification, and quality conformance by device class are specified in table II.

3.7 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2. At the option of the manufacturer, the following marking may be omitted from the body of the microcircuit, but shall be retained on the initial container:

(a) Country of origin.

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirement	Subgroups (see table III)		
	Class A devices	Class B devices	Class C devices
Interim electrical parameters (pre burn-in) (Method 5004)	1	1	None
Final electrical test parameters (Method 5004)	1*, 2, 3, 9	1*, 2, 3, 9	1
Group A test requirements (Method 5005)	1, 2, 3, 9, 10, 11	1, 2, 3, 9	1, 2, 3, 9
Groups B and C end point electrical parameters (Method 5005)	1, 2, 3	1, 2, 3	1
Additional electrical subgroups for group C periodic inspections	None	10, 11	None

4. PRODUCT ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and Method 5005 of MIL-STD-883, except as modified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, and C inspections (see 4.4.1, 4.4.2, and 4.4.3). After qualification of one or more electrically and structurally similar types with a single lead finish, other lead finishes of the same case outline may be qualified by submitting a single type in the qualified case outline to the group B, subgroup 3 test and the group C, subgroups 1, 3, and 4 tests.

4.3 Screening. Screening shall be in accordance with Method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

(a) Test samples for the group B bond strength test specified in Method 5005 of MIL-STD-883 may, at the manufacturer's option, be randomly selected immediately following the internal visual (precap) inspection and prior to sealing (see 4.4.2(b)).
(b) Temperature cycling (Method 1010 of MIL-STD-883).
    (1) Omit seal test as post-test measurement.
(c) Thermal shock (Method 1011 of MIL-STD-883), when substituted for temperature cycling.
    (1) Omit seal test as post-test measurement.
(d) Burn-in test (Method 1015 of MIL-STD-883).
    (1) Test condition D or E, using the circuit shown on figure 4, or equivalent.
    (2) T_A = 125° C minimum.
(e) Reverse bias burn-in and interim electrical test in accordance with 3.1.10 of Method 5004 of MIL-STD-883 may be omitted.
(f) Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.
(g) External visual inspection shall not include measurement of case and lead dimensions.
(h) Percent defective allowable (PDA) - The PDA is specified as 5 percent for class A devices and 10 percent for class B devices based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with Method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA.

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The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510.

4.4.1 Group A inspection. Group A inspection shall consist of the test subgroups and LTPD values shown in table I of Method 5005 of MIL-STD-883 and as follows:

(a) Tests shall be as specified in table II.
(b) Subgroups 4, 5, 6, 7, and 8 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall consist of the test subgroups and LTPD values shown in table II of Method 5005 of MIL-STD-883 and as follows:

(a) End point electrical parameters shall be as specified in table II.
(b) Bond strength test may be conducted on samples collected prior to sealing (see 4.3(a)).

4.4.3 Group C inspection. Group C inspection shall consist of the test subgroups and LTPD values shown in table III of Method 5005 of MIL-STD-883 and as follows:

(a) End point electrical parameters shall be as specified in table II.
(b) Subgroups 7 and 8 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.
(c) Lead bend in only one direction is required for initial conditioning prior to moisture resistance and salt atmosphere tests.
(d) High temperature storage test (Method 1008 of MIL-STD-883) conditions:
    (1) Temperature: 150 +10° C.
    (2) Duration: 1,000 hours, except as otherwise permitted by Appendix B of MIL-M-38510.
(e) Operating life test (Method 1005 of MIL-STD-883) conditions:
    (1) Test condition D or E, using the circuit shown on figure 4, or equivalent.
    (2) T_A 125° C, minimum.
    (3) Test duration: 1,000 hours, except as permitted by Appendix B of MIL-M-38510.
(f) Omit steady state reverse bias test.

4.5 Methods of examination and test. Methods of examination and test shall be as specified in the appropriate tables and as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

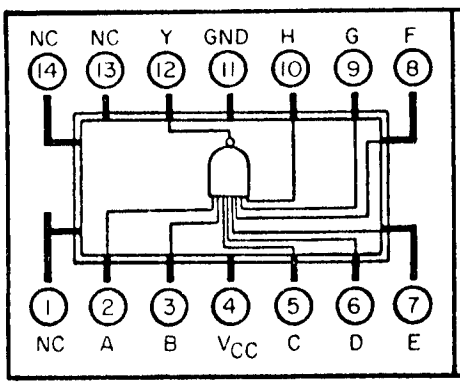
4.5.2 Life test cooldown procedure. When devices are measured at 25° C following application of the operating life or burn-in test condition, they shall be cooled to room temperature prior to removal of the bias. Alternately, the bias may be removed during cooling if the case temperature is reduced to room temperature within 30 minutes after removal of the test condition.

4.6 Inspection of preparation for delivery. Inspection of preparation for delivery shall be in accordance with MIL-M-38510, except that the rough handling test shall not apply.

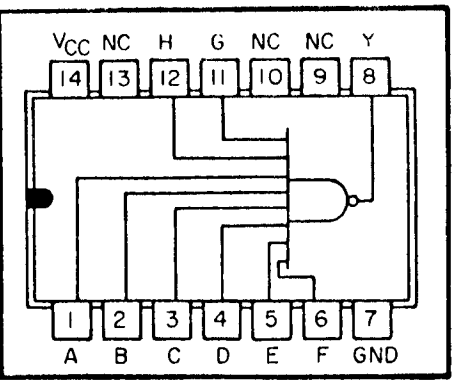
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Device type 01

Case A, B, and D

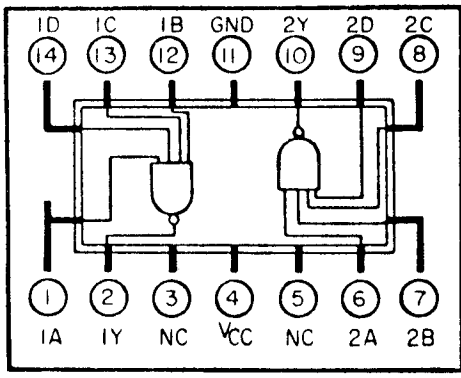


Case C

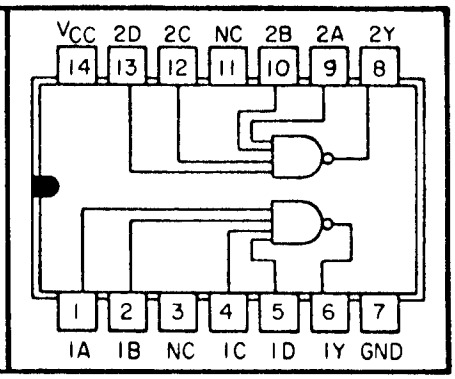


Device type 02

Case A, B, and D

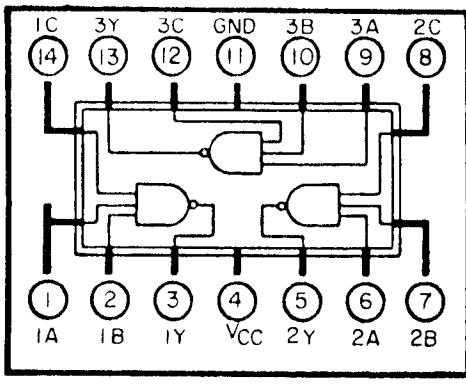


Case C



Device type 03

Case A, B, and D



Case C

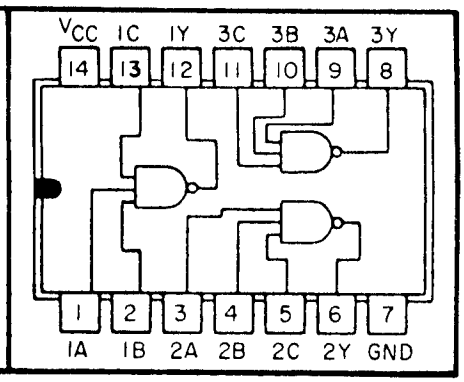
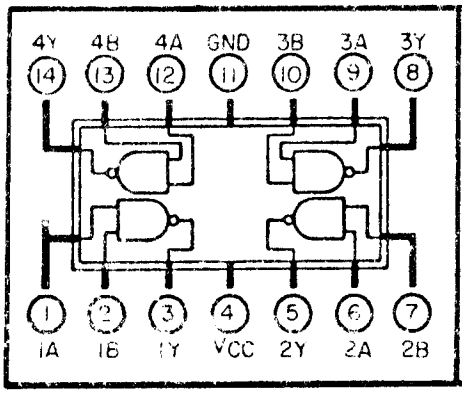


FIGURE 1. Logic diagrams and terminal connections (Top Views)

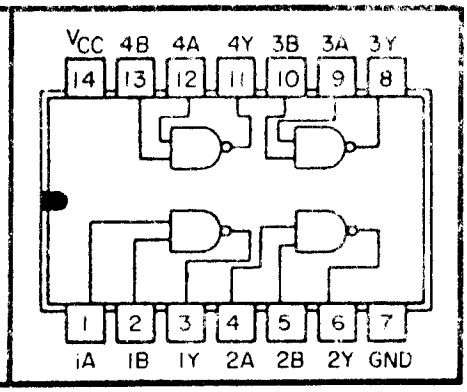
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* Device types 04 and 06

Case A, B, and D



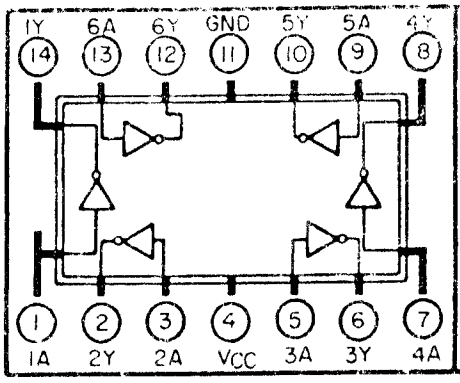
Case C



* Device type 06 in accordance with "Case C" only.

Device type 05

Case A, B, and D



Case C

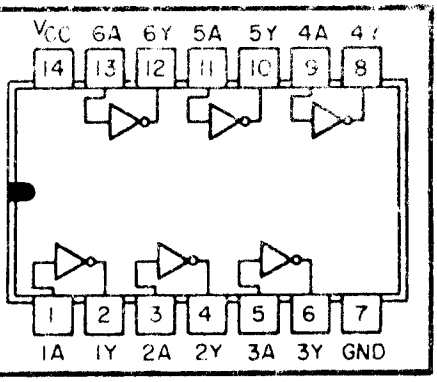


FIGURE 1. Logic diagram and terminal connections (Top Views) - Continued.

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Device type 01

Truth table								
Input								Output
A	B	C	D	E	F	G	H	Y
H	H	H	H	H	H	H	H	L
All other combinations of H and L at the inputs give H output.								

Positive logic $Y = \overline{ABCDEFGH}$

Device type 05

Truth table each gate	
Input	Output
A	Y
L	H
H	L

Positive logic $Y = \overline{A}$

Device type 03

Truth table			
Input			Output
A	B	C	Y
L	L	L	H
H	L	L	H
L	H	L	H
H	H	L	H
L	L	H	H
H	L	H	H
L	H	H	H
H	H	H	L

Positive logic $Y = \overline{ABC}$

Device type 02

Truth table				
Input				Output
A	B	C	D	Y
L	L	L	L	H
H	L	L	L	H
L	H	L	L	H
H	H	L	L	H
L	L	H	L	H
H	L	H	L	H
L	H	H	L	H
H	H	H	L	H
L	L	L	H	H
H	L	L	H	H
L	H	L	H	H
H	H	L	H	H
L	L	H	H	H
H	L	H	H	H
L	H	H	H	H
H	H	H	H	L

Positive logic $Y = \overline{ABCD}$

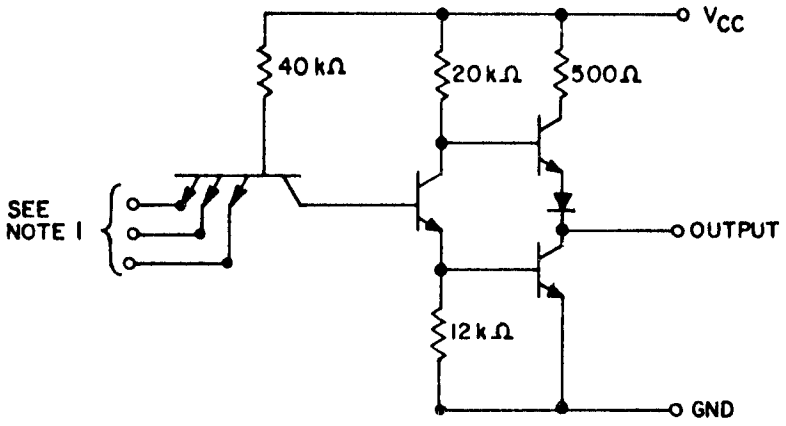
Device types 04 and 06

Truth table each gate		
Input		Output
A	B	Y
L	L	H
H	L	H
L	H	H
H	H	L

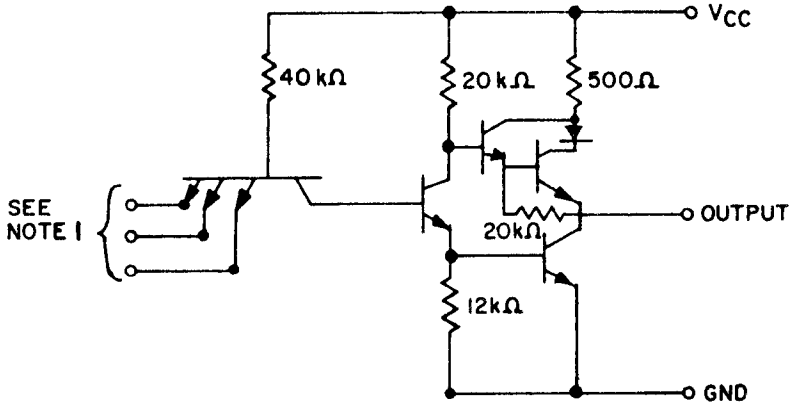
Positive logic $Y = \overline{AB}$

FIGURE 2. Truth tables and logic equations.

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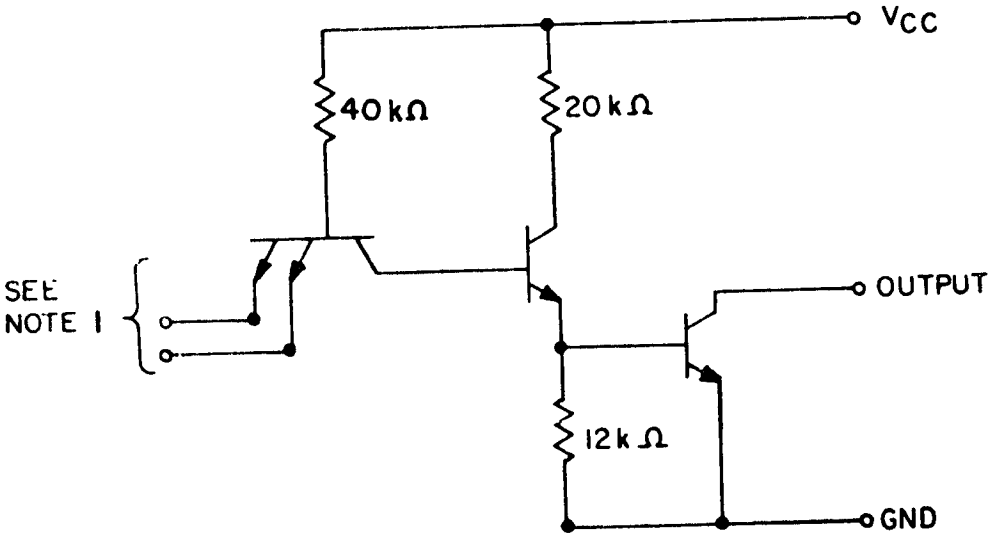
SCHEMATIC A



SCHEMATIC B

FIGURE 3. Schematic circuit for device types 01, 02, 03, 04 and 05.

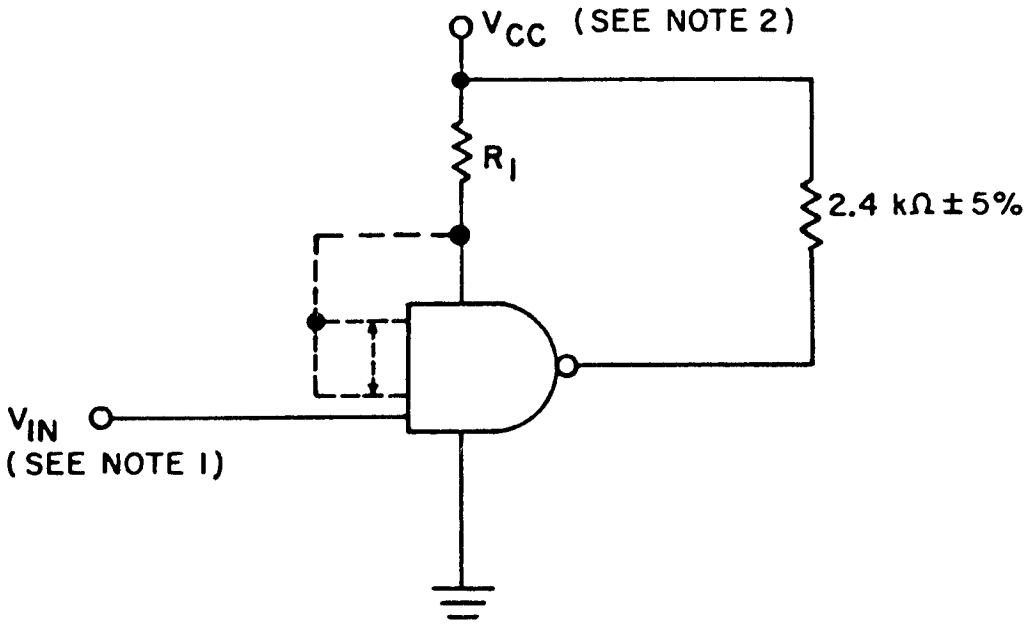
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NOTES:
1. Number of inputs is determined by logic configuration shown in figure 1.
2. All resistance values shown are nominal.

FIGURE 3. Schematic circuit for device type 06-Continued.

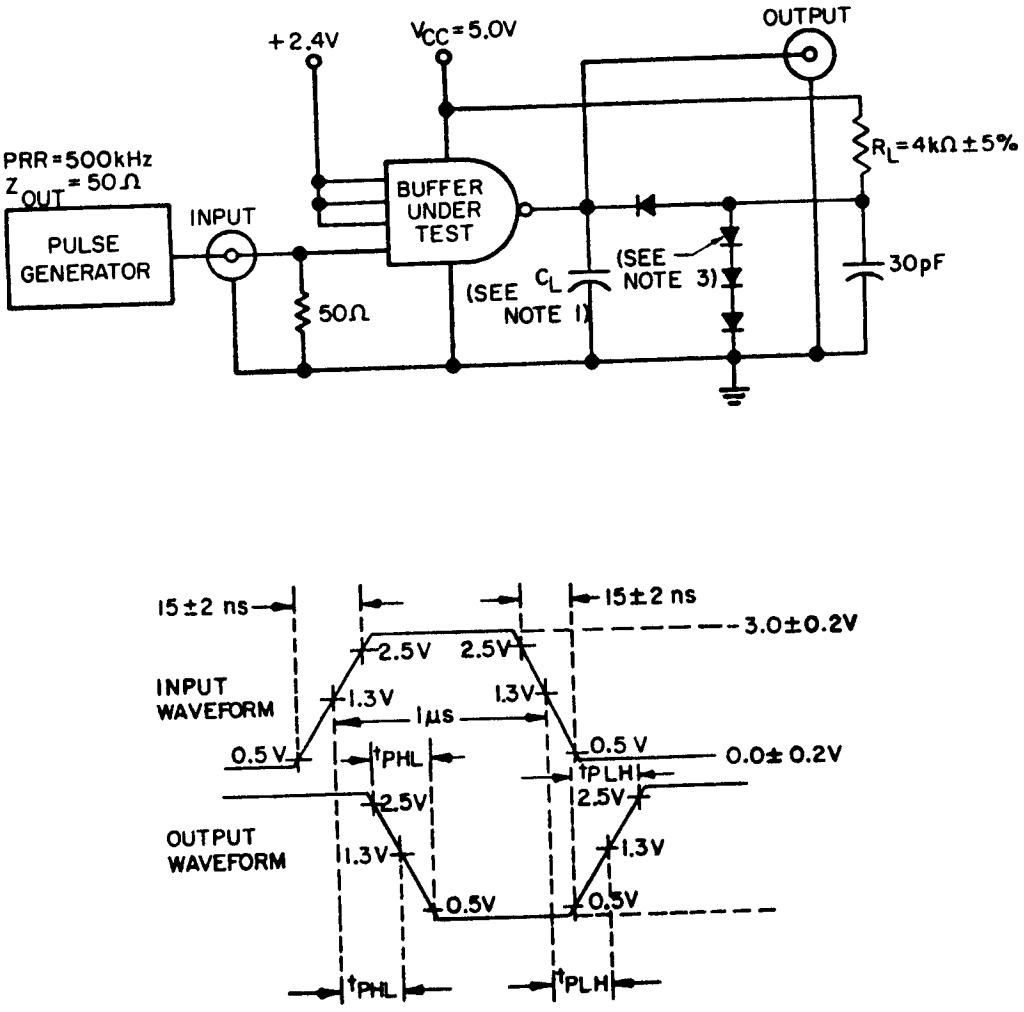
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NOTES:
1. One input of each gate shall be driven. The remaining inputs of each gate shall be connected to V_{CC} through the appropriate current limiting resistors.
2. V_{CC} and R_1 are such that the minimum voltage at the device terminals is 5 volts.
3. If parallel excitation is used, the clock frequency shall be 100kHz with V_{IN} = 3 volts minimum and minimum duty cycle of 50%.

FIGURE 4. Burn-in and life test circuit.

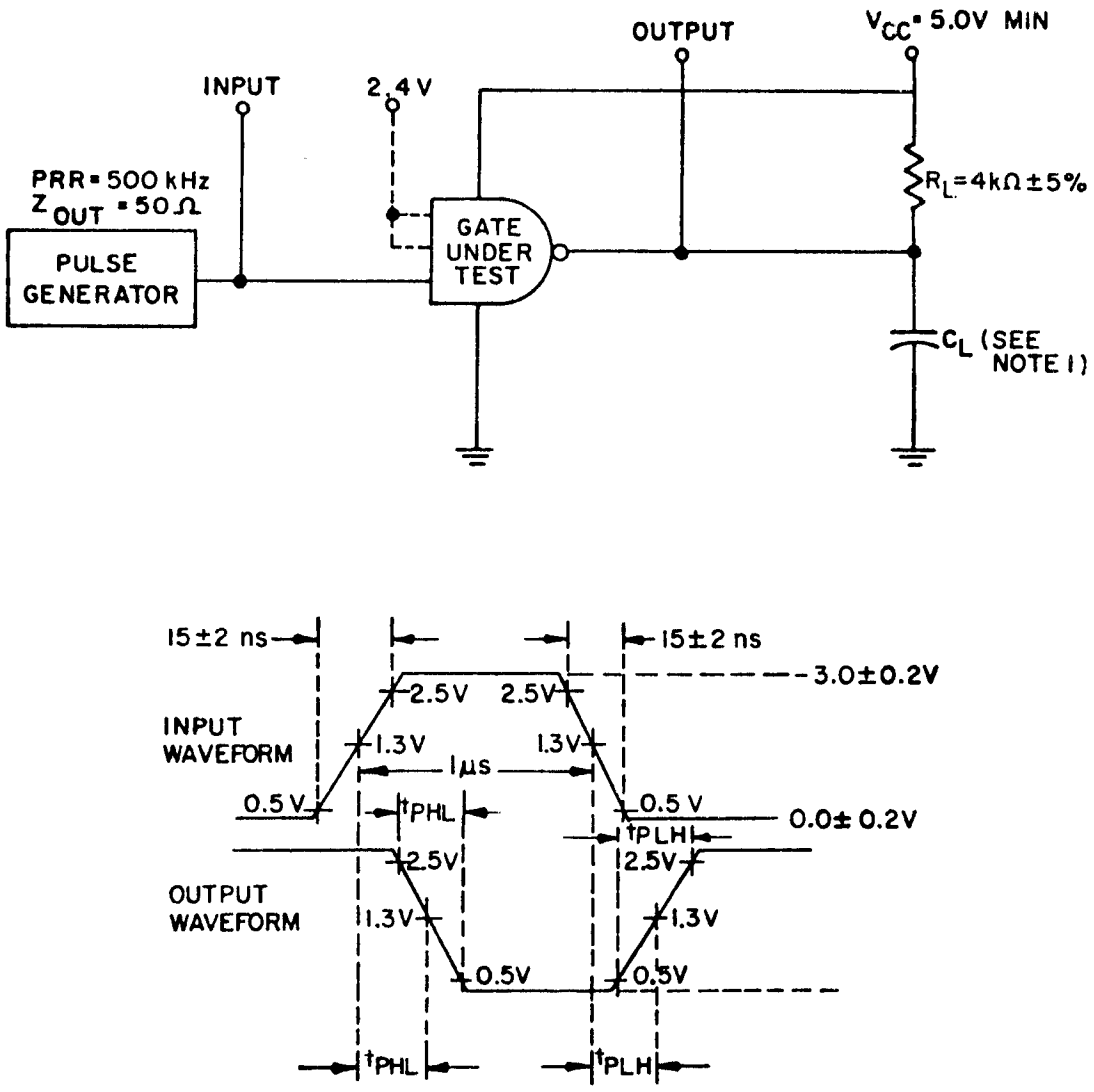
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NOTES:
1. C_L=50pF minimum, including scope probe, wiring, and stray capacitance, without package in test fixture.
2. Voltage measurements are to be made with respect to network ground terminal.
3. All diodes are 1N3064 or equivalent.

FIGURE 5. Switching time test circuit except for open collector circuits (types 01, 02, 03, 04, and 05).

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NOTES:
1. C_L = 50pF minimum, including scope probe, wiring, and stray capacitance, without package in test fixture.
2. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 6. Switching time test circuit (type 06).

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
			Case C	13	1	2	14	3	4	5	6	11	12	7	8	9	10		Min	Max	Unit
			Test No.	NC	A	B	VCC	C	D	E	F	G	H	GND	Y	NC	NC				
1	V_{OL}	3007	1		2.0 v	2.0 v	4.5 v	2.0 v	2.0 v	2.0 v	2.0 v	2.0 v	2.0 v	GND	2 mA			Y		0.3	V
T_A = 25°C	V_{OH}	3006	2		0.7 v	5.5 v		5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	5.5 v		-100 µA			Y	2.4		
			3		5.5 v	0.7 v		5.5 v										Y			
			4		5.5 v	5.5 v		0.7 v										Y			
			5					5.5 v	0.7 v									Y			
			6						5.5 v	0.7 v								Y			
			7							5.5 v	0.7 v							Y			
			8								5.5 v	0.7 v						Y			
			9								5.5 v	5.5 v	0.7 v					Y			
	I_{OS}	3011	10		GND	GND	5.5 v	GND	GND	GND	GND	GND	GND		GND			Y	-3	-15	mA
	I_{IH1}	3010	11		2.4 v	GND		GND	GND	GND	GND	GND	GND					A		10	µA
			12		GND	2.4 v		GND										B			
			13			GND		2.4 v	2.4 v	2.4 v	2.4 v	2.4 v	2.4 v					C			
			14					GND	GND	GND	GND	GND						D			
			15															E			
			16															F			
			17															G			
			18															H			
	I_{IH2}		19		5.5 v	5.5 v							GND					A		100	
			20		GND	GND												B			
			21					5.5 v	5.5 v	5.5 v	5.5 v	5.5 v						C			
			22					GND	GND	GND	GND	GND						D			
			23								5.5 v	5.5 v						E			
			24								GND	GND						F			
			25										5.5 v					G			
			26										5.5 v					H			
	I_{IL}	3009	27		0.3 v	5.5 v		5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	5.5 v					A	-60	-180	
			28		5.5 v	0.3 v		5.5 v										B			
			29			5.5 v		0.3 v										C			
			30					5.5 v	0.3 v									D			
			31						5.5 v	0.3 v								E			
			32							5.5 v	0.3 v							F			
			33								5.5 v	0.3 v						G			
			34								5.5 v	5.5 v	0.3 v					H			
	I_{CCL}	3005	35		5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	5.5 v					VCC		0.51	mA
	I_{CCH}	3005	36		GND	GND	5.5 v	GND	GND	GND	GND	GND	GND					VCC		0.33	mA
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125°C.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55°C.																				
9	t_{PHL}	3003	37 CKT A		IN	2.4 v	5.0 v	2.4 v	2.4 v	2.4 v	2.4 v	2.4 v	2.4 v	GND	OUT			A to Y	3	90	ns
T_A = 25°C	t_{PHL}	(Fig. 5)	37 CKT B															A to Y		80	
	t_{PLH}		38 CKT A															A to Y		60	
	t_{PLH}		38 CKT B															A to Y		60	
10	t_{PHL}		39 CKT A		IN	2.4 v	5.0 v	2.4 v	2.4 v	2.4 v	2.4 v	2.4 v	2.4 v	GND	OUT			A to Y		140	
T_A = 125°C	t_{PHL}		39 CKT B															A to Y		90	
	t_{PLH}		40 CKT A															A to Y		90	
	t_{PLH}		40 CKT B															A to Y		90	
11	Same tests, terminal conditions and limits as for subgroup 10, except T_A = -55°C.																				

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TABLE III. Group A inspection for device type 02.
Terminal condition (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas terminal	Test limits		
			Case C	1	6	3	14	11	9	10	12	13	8	7	2	4	5				
			Test No.	1A	1Y	NC	VCC	NC	2A	2B	2C	2D	2Y	GND	1B	1C	1D		Min	Max	Unit
1 T_A=25° C	V_{OL}	3007	1	2.0 v	2 mA		4.5 v		5.5 v	5.5 v	5.5 v	5.5 v		GND	2.0 v	2.0 v	2.0 v	1Y		.3	V
	V_{OL}	3007	2	5.5 v					2.0 v	2.0 v	2.0 v	2.0 v	2 mA		5.5 v	5.5 v	5.5 v	2Y		.3	
	V_{OH}	3008	3	0.7 v	-100 µA				5.5 v	5.5 v	5.5 v	5.5 v			5.5 v			1Y	2.4		
			4	5.5 v											0.7 v			1Y			
			5												5.5 v	0.7 v		1Y			
			6													5.5 v	0.7 v	1Y			
			7						0.7 v				-100 µA				5.5 v	2Y			
			8						5.5 v	0.7 v								2Y			
			9							5.5 v	0.7 v							2Y			
			10							5.5 v	5.5 v	0.7 v						2Y			
	I_{OS}	3011	11	GND	GND		5.5 v								GND	GND	GND	1Y	-3	-15	mA
	I_{OS}	3011	12						GND	GND	GND	GND	GND					2Y	-3	-15	mA
	I_{IH1}	3010	13	2.4 v											GND	GND	GND	1A		10	µA
			14	GND											2.4 v	GND		1B			
			15												GND	2.4 v		1C			
			16													GND	2.4 v	1D			
			17						2.4 v								GND	2A			
			18						GND	2.4 v								2B			
			19							GND	2.4 v							2C			
			20								GND	2.4 v						2D			
	I_{IH2}		21	5.5 v								GND						1A		100	
			22	GND											5.5 v			1B			
			23												GND	5.5 v		1C			
			24													GND	5.5 v	1D			
			25						5.5 v								GND	2A			
			26						GND	5.5 v								2B			
			27							GND	5.5 v							2C			
			28								GND	5.5 v						2D			
	I_{IL}	3009	29	0.3 v					5.5 v	5.5 v	5.5 v				5.5 v	5.5 v	5.5 v	1A	-60	-180	
			30	5.5 v											0.3 v	5.5 v		1B			
			31												5.5 v	0.3 v		1C			
			32													5.5 v	0.3 v	1D			
			33						0.3 v								5.5 v	2A			
			34						5.5 v	0.3 v								2B			
			35							5.5 v	0.3 v							2C			
			36								5.5 v	0.3 v						2D			
	I_{CCL}	3005	37								5.5 v	5.5 v						VCC		1.02	mA
	I_{CCH}	3005	38	GND					GND	GND	GND	GND			GND	GND	GND	VCC		0.40	mA
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125° C.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55° C.																				

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
			Case C	1	6	3	14	11	9	10	12	13	8	7	2	4	5		Min	Max	Unit
			Test No.	1A	1Y	NC	VCC	NC	2A	2B	2C	2D	2Y	GND	1B	1C	1D				
9 T_A=25° C	t_{PHL}	3003 (Fig. 5)	39 CKT A	IN	OUT		5.0 v							GND	2.4 v	2.4 v	2.4 v	1A to 1Y	3	60	ns
			39 CKT B	IN	OUT										2.4 v	2.4 v	2.4 v	1A to 1Y		50	
			40 CKT A						IN	2.4 v	2.4 v	2.4 v	OUT					2A to 2Y		60	
			40 CKT B						IN	2.4 v	2.4 v	2.4 v	OUT					2A to 2Y		50	
	t_{PLH}		41 CKT A	IN	OUT										2.4 v	2.4 v	2.4 v	1A to 1Y		60	
			41 CKT B	IN	OUT										2.4 v	2.4 v	2.4 v	1A to 1Y		45	
			42 CKT A						IN	2.4 v	2.4 v	2.4 v	OUT					2A to 2Y		60	
			42 CKT B						IN	2.4 v	2.4 v	2.4 v	OUT					2A to 2Y		45	
10 T_A=125°C	t_{PHL}		43 CKT A	IN	OUT										2.4 v	2.4 v	2.4 v	1A to 1Y		90	
			43 CKT B	IN	OUT										2.4 v	2.4 v	2.4 v	1A to 1Y		60	
			44 CKT A						IN	2.4 v	2.4 v	2.4 v	OUT					2A to 2Y		90	
			44 CKT B						IN	2.4 v	2.4 v	2.4 v	OUT					2A to 2Y		60	
	t_{PLH}		45 CKT A	IN	OUT										2.4 v	2.4 v	2.4 v	1A to 1Y		90	
			45 CKT B	IN	OUT										2.4 v	2.4 v	2.4 v	1A to 1Y		55	
			46 CKT A						IN	2.4 v	2.4 v	2.4 v	OUT					2A to 2Y		90	
			46 CKT B						IN	2.4 v	2.4 v	2.4 v	OUT					2A to 2Y		55	
11	Same tests, terminal conditions and limits as for subgroup 10, except T_A = -55° C.																				

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D Case C Test No.	1 1A	2 1B	3 12 1Y	4 14 VCC	5 6 2Y	6 3 2A	7 4 2B	8 5 2C	9 9 3A	10 10 3B	11 7 GND	12 11 3C	13 8 3Y	14 13 1C	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
1 $T_A = 25°C$	V_{OL}	3007	1	2.0 v	2.0 v	2 mA	4.5 v	2 mA	5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	GND	5.5 v		2.0 v	1Y		0.3	V
			2	5.5 v	5.5 v				2.0 v	2.0 v	2.0 v	5.5 v	5.5 v		5.5 v		5.5 v	2Y			
			3	5.5 v					5.5 v	5.5 v	5.5 v	2.0 v	2.0 v		2.0 v	2 mA		3Y			
	V_{OH}	3006	4	0.7 v	-100μA							5.5 v	5.5 v		5.5 v			1Y	2.4		
			5	5.5 v	0.7 v													1Y			
			6		5.5 v			-100μA	0.7 v	0.7 v	0.7 v						0.7 v	1Y			
			7						5.5 v	5.5 v	5.5 v						5.5 v	2Y			
			8													-100μA		2Y			
			9									0.7 v						3Y			
			10									5.5 v	0.7 v					3Y			
			11									5.5 v	5.5 v					3Y			
			12																		
	I_{OS}	3011	13	GND	GND	GND	5.5 v	GND	GND	GND	GND	GND	GND		GND	GND	GND	1Y	-3	-15	mA
			14															2Y			
			15															3Y			
	I_{IH1}	3010	16	2.4 v	GND	GND			GND	GND	GND	GND	GND				GND	1A		10	μA
			17	GND	2.4 v	2.4 v											GND	1B			
			18		GND	GND			2.4 v	2.4 v	2.4 v	2.4 v	2.4 v				2.4 v	1C			
			19						GND	GND	GND	GND	GND				GND	2A			
			20															2B			
			21															2C			
			22															3A			
			23															3B			
			24															3C			
	I_{IH2}		25	5.5 v	5.5 v				5.5 v	5.5 v	5.5 v	5.5 v	5.5 v				5.5 v	1A		100	
			26	GND	5.5 v													1B			
			27		GND													1C			
			28						5.5 v	5.5 v	5.5 v	5.5 v	5.5 v				5.5 v	2A			
			29						GND	GND	GND	GND	GND				GND	2B			
			30															2C			
			31															3A			
			32															3B			
			33															3C			
	I_{IL}	3009	34	0.3 v	5.5 v				5.5 v	5.5 v	5.5 v	5.5 v	5.5 v				5.5 v	1A	-60	-180	
			35	5.5 v	0.3 v												5.5 v	1B			
			36		5.5 v				0.3 v	0.3 v	0.3 v						0.3 v	1C			
			37						5.5 v	5.5 v	5.5 v						5.5 v	2A			
			38															2B			
			39									0.3 v	0.3 v					2C			
			40									5.5 v	5.5 v					3A			
			41									5.5 v	5.5 v					3B			
			42															3C			
	I_{CCH}	3005	43	GND	GND	GND			GND	GND	GND	GND	GND		GND		GND	VCC		0.60	mA
	I_{CCL}	3005	44	5.5 v	5.5 v				5.5 v	5.5 v	5.5 v	5.5 v	5.5 v		5.5 v		5.5 v	VCC		1.53	mA
2	Same tests, terminal conditions and limits as for subgroup 1, except $T_A = 125°C$.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except $T_A = -55°C$.																				

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
			Case C	1	2	12	14	6	3	4	5	9	10	7	11	8	13		Min	Max	Unit
			Test No.	1A	1B	1Y	VCC	2Y	2A	2B	2C	3A	3B	GND	3C	3Y	1C				
9 T_A=25°C	t_{PHL}	3003 (Fig. 5)	45 CKT A	IN	2.4 v	OUT	5.0 v							GND			2.4 v	1A to 1Y	3	60	ns
			45 CKT B	IN	2.4 v	OUT											2.4 v	1A to 1Y		50	
			46 CKT A					OUT	IN	2.4 v	2.4 v							2A to 2Y		60	
			46 CKT B					OUT	IN	2.4 v	2.4 v							2A to 2Y		50	
			47 CKT A									IN	2.4 v		2.4 v	OUT		3A to 3Y		60	
			47 CKT B									IN	2.4 v		2.4 v	OUT		3A to 3Y		50	
	t_{PLH}		48 CKT A	IN	2.4 v	OUT											2.4 v	1A to 1Y		60	
			48 CKT B	IN	2.4 v	OUT											2.4 v	1A to 1Y		45	
			49 CKT A					OUT	IN	2.4 v	2.4 v							2A to 2Y		60	
			49 CKT B					OUT	IN	2.4 v	2.4 v							2A to 2Y		45	
			50 CKT A									IN	2.4 v		2.4 v	OUT		3A to 3Y		60	
			50 CKT B									IN	2.4 v		2.4 v	OUT		3A to 3Y		45	
10 T_A=125°C	t_{PHL}		51 CKT A	IN	2.4 v	OUT											2.4 v	1A to 1Y		90	
			51 CKT B	IN	2.4 v	OUT											2.4 v	1A to 1Y		60	
			52 CKT A					OUT	IN	2.4 v	2.4 v							2A to 2Y		90	
			52 CKT B					OUT	IN	2.4 v	2.4 v							2A to 2Y		60	
			53 CKT A									IN	2.4 v		2.4 v	OUT		3A to 3Y		90	
			53 CKT B									IN	2.4 v		2.4 v	OUT		3A to 3Y		60	
	t_{PLH}		54 CKT A	IN	2.4 v	OUT											2.4 v	1A to 1Y		90	
			54 CKT B	IN	2.4 v	OUT											2.4 v	1A to 1Y		55	
			55 CKT A					OUT	IN	2.4 v	2.4 v							2A to 2Y		90	
			55 CKT B					OUT	IN	2.4 v	2.4 v							2A to 2Y		55	
			56 CKT A									IN	2.4 v		2.4 v	OUT		3A to 3Y		90	
			56 CKT B									IN	2.4 v		2.4 v	OUT		3A to 3Y		55	
11	Same tests, terminal conditions and limits as for subgroup 10, except T_A = -55° C.																				

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TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	Case C	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits Min	Test limits Max	Unit
			Test No.		1A	1B	1Y	VCC	2Y	2A	2B	3Y	3A	3B	GND	4A	4B	4Y				
1 T_A = 25° C	V_{OL}	3007	1		2.0 v	2.0 v	2 mA	4.5 v	2 mA	5.5 v	5.5 v	2 mA	5.5 v	5.5 v	GND	5.5 v	5.5 v	2 mA	1Y		0.3	V
			2		5.5 v	5.5 v				2.0 v	2.0 v		5.5 v	5.5 v		2.0 v	2.0 v		2Y			
			3							5.5 v	5.5 v		2.0 v	2.0 v					3Y			
			4										5.5 v	5.5 v					4Y			
	V_{OH}	3006	5		0.7 v	0.7 v	-100 µA		-100 µA	0.7 v	0.7 v	-100 µA	0.7 v	0.7 v		5.5 v	5.5 v	-100µA	1Y	2.4		
			6		5.5 v	5.5 v	-100 µA		-100 µA	5.5 v	5.5 v	-100 µA	5.5 v	5.5 v				-100µA	1Y			
			7													0.7 v	0.7 v		2Y			
			8													5.5 v	5.5 v		2Y			
			9																3Y			
			10																3Y			
			11																4Y			
			12																4Y			
	I_{OS}	3011	13		GND	GND	GND	5.5 v	GND	GND	GND	GND	GND	GND		GND	GND	GND	1Y	-3	-15	mA
			14																2Y			
			15																3Y			
			16																4Y			
	I_{IH1}	3010	17		2.4 v	2.4 v				GND	GND		GND	GND		GND	GND		1A		10	µA
			18		GND	GND				GND									1B			
			19							2.4 v	2.4 v		2.4 v	2.4 v					2A			
			20							GND	GND		GND	GND					2B			
			21																3A			
			22																3B			
			23																4A			
			24																4B			
	I_{IH2}		25		5.5 v	5.5 v				5.5 v	5.5 v		5.5 v	5.5 v		5.5 v	GND		1A		100	
			26		GND	GND				GND									1B			
			27								5.5 v		5.5 v	5.5 v					2A			
			28								GND		GND	GND					2B			
			29													5.5 v	5.5 v		3A			
			30													GND	GND		3B			
			31																4A			
			32																4B			
	I_{IL}	3009	33		0.3 v	5.5 v				5.5 v	5.5 v		5.5 v	5.5 v		5.5 v	5.5 v		1A	-60	-180	
			34		5.5 v	0.3 v				5.5 v	5.5 v								1B			
			35			5.5 v				0.3 v	5.5 v		0.3 v	0.3 v					2A			
			36							5.5 v	5.5 v		GND	GND		0.3 v	0.3 v		2B			
			37																3A			
			38																3B			
			39																4A			
			40																4B			
	I_{CCH}	3005	41		GND	GND	GND			GND	GND		GND	GND		GND	GND		VCC		0.80	mA
	I_{CCL}	3005	42		5.5 v	5.5 v				5.5 v	5.5 v		5.5 v	5.5 v		5.5 v	5.5 v		VCC		2.04	mA
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125° C.																					
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55° C.																					

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
			Case C	1	2	3	14	6	4	5	8	9	10	7	12	13	11		Min	Max	Unit
			Test No.	1A	1B	1Y	VCC	2Y	2A	2B	3Y	3A	3B	GND	4A	4B	4Y				
9 T_A=25° C	t_{PHL}	3003 (Fig. 5)	43 CKT A	IN	2.4 v	OUT	5.0 v							GND				1A to 1Y	3	60	ns
			43 CKT B	IN	2.4 v	OUT												1A to 1Y		50	
			44 CKT A					OUT	IN	2.4 v								2A to 2Y		60	
			44 CKT B					OUT	IN	2.4 v								2A to 2Y		50	
			45 CKT A								OUT	IN	2.4 v					3A to 3Y		60	
			45 CKT B								OUT	IN	2.4 v					3A to 3Y		50	
			46 CKT A												IN	2.4 v	OUT	4A to 4Y		60	
			46 CKT B												IN	2.4 v	OUT	4A to 4Y		50	
	t_{PLH}		47 CKT A	IN	2.4 v	OUT												1A to 1Y		60	
			47 CKT B	IN	2.4 v	OUT												1A to 1Y		45	
			48 CKT A					OUT	IN	2.4 v								2A to 2Y		60	
			48 CKT B					OUT	IN	2.4 v								2A to 2Y		45	
			49 CKT A								OUT	IN	2.4 v					3A to 3Y		60	
			49 CKT B								OUT	IN	2.4 v					3A to 3Y		45	
			50 CKT A												IN	2.4 v	OUT	4A to 4Y		60	
			50 CKT B												IN	2.4 v	OUT	4A to 4Y		45	
10 T_A=125°C	t_{PHL}		51 CKT A	IN	2.4 v	OUT												1A to 1Y		90	
			51 CKT B	IN	2.4 v	OUT												1A to 1Y		60	
			52 CKT A					OUT	IN	2.4 v								2A to 2Y		90	
			52 CKT B					OUT	IN	2.4 v								2A to 2Y		60	
			53 CKT A								OUT	IN	2.4 v					3A to 3Y		90	
			53 CKT B								OUT	IN	2.4 v					3A to 3Y		60	
			54 CKT A												IN	2.4 v	OUT	4A to 4Y		90	
			54 CKT B												IN	2.4 v	OUT	4A to 4Y		60	
	t_{PLH}		55 CKT A	IN	2.4 v	OUT												1A to 1Y		90	
			55 CKT B	IN	2.4 v	OUT												1A to 1Y		55	
			56 CKT A					OUT	IN	2.4 v								2A to 2Y		90	
			56 CKT B					OUT	IN	2.4 v								2A to 2Y		55	
			57 CKT A								OUT	IN	2.4 v					3A to 3Y		90	
			57 CKT B								OUT	IN	2.4 v					3A to 3Y		55	
			58 CKT A												IN	2.4 v	OUT	4A to 4Y		90	
			58 CKT B												IN	2.4 v	OUT	4A to 4Y		55	
11	Same tests, terminal conditions and limits as for subgroup 10, except T_A = -55° C.																				

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TABLE III. Group A inspection for device type 05.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 1 / 1A	2 / 4 / 2Y	3 / 3 / 2A	4 / 14 / VCC	5 / 5 / 3A	6 / 6 / 3Y	7 / 9 / 4A	8 / 8 / 4Y	9 / 11 / 5A	10 / 10 / 5Y	11 / 7 / GND	12 / 12 / 6Y	13 / 13 / 6A	14 / 2 / 1Y	Meas. terminal	Test limits Min	Max	Unit
1 T_A = 25°C	V_{OL}	3007	1	2.0 v	2 mA	5.5 v	4.5 v	5.5 v		5.5 v		5.5 v		GND		5.5 v	2 mA	1Y		0.3	V
			2	5.5 v		2.0 v		5.5 v										2Y			
			3			5.5 v		2.0 v	2 mA									3Y			
			4					5.5 v		2.0 v	2 mA							4Y			
			5							5.5 v		2.0 v	2 mA					5Y			
			6									5.5 v			2 mA	2.0 v		6Y			
	V_{OH}	3006	7	0.7 v	-100 µA	0.7 v		0.7 v	-100 µA	0.7 v	-100 µA	0.7 v	-100 µA			5.5 v	-100 µA	1Y	2.4		
			8	5.5 v		5.5 v		5.5 v		5.5 v		5.5 v			-100 µA	0.7 v		2Y			
			9							5.5 v								3Y			
			10															4Y			
			11															5Y			
			12															6Y			
	I_{OS}	3011	13	GND	GND	GND	5.5 v	GND	GND	GND	GND	GND	GND		GND		GND	1Y	-3	-15	mA
			14															2Y			
			15															3Y			
			16															4Y			
			17															5Y			
			18															6Y			
	I_{IH1}	3010	19	2.4 v		GND		GND		GND		GND						1A		10	µA
			20	GND		2.4 v		GND										2A			
			21			GND		2.4 v		2.4 v		2.4 v						3A			
			22					GND		GND		GND						4A			
			23															5A			
			24													2.4 v		6A			
	I_{IH2}		25	5.5 v		5.5 v		5.5 v		5.5 v		5.5 v				GND		1A		100	
			26	GND		GND		GND										2A			
			27															3A			
			28					5.5 v		GND		GND						4A			
			29					GND										5A			
			30													5.5 v		6A			
	I_{IL}	3009	31	0.3 v		5.5 v		5.5 v		5.5 v		5.5 v						1A	-60	-180	
			32	5.5 v		0.3 v		5.5 v										2A			
			33			5.5 v		0.3 v		0.3 v		0.3 v						3A			
			34					5.5 v		5.5 v								4A			
			35									0.3 v						5A			
			36									5.5 v				0.3 v		6A			
	I_{CCL}	3005	37									5.5 v				5.5 v		VCC		3.10	mA
	I_{CCH}	3005	38	GND		GND		GND		GND		GND				GND		VCC		1.20	mA
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125° C.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55° C.																				

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas.	Test limits	
			Case C	1	4	3	14	5	6	9	8	11	16	7	12	13	2			
			Test No.	1A	2Y	2A	VCC	3A	3Y	4A	4Y	5A	5Y	GND	6Y	6A	1Y	terminal	Min	Max	Unit
9 T_A=25° C	t_{PHL}	3003 (Fig. 5)	39 CKT A	IN			5.0 V										OUT	1A to 1Y	3	60	ns
			39 CKT B	IN													OUT	1A to 1Y		50	
			40 CKT A		OUT	IN		IN										2A to 2Y		60	
			40 CKT B		OUT	IN		IN										2A to 2Y		50	
			41 CKT A						OUT				OUT					3A to 3Y		60	
			41 CKT B						OUT				OUT					3A to 3Y		50	
			42 CKT A							IN	OUT							4A to 4Y		60	
			42 CKT B							IN	OUT							4A to 4Y		50	
			43 CKT A									IN	OUT					5A to 5Y		60	
			43 CKT B									IN	OUT					5A to 5Y		50	
			44 CKT A												OUT	IN		6A to 6Y		60	
			44 CKT B												OUT	IN		6A to 6Y		50	
	t_{PLH}		45 CKT A	IN													OUT	1A to 1Y		60	
			45 CKT B	IN													OUT	1A to 1Y		45	
			46 CKT A		OUT	IN		IN										2A to 2Y		60	
			46 CKT B		OUT	IN		IN										2A to 2Y		45	
			47 CKT A						OUT				OUT					3A to 3Y		60	
			47 CKT B						OUT				OUT					3A to 3Y		45	
			48 CKT A							IN	OUT							4A to 4Y		60	
			48 CKT B							IN	OUT							4A to 4Y		45	
			49 CKT A									IN	OUT					5A to 5Y		60	
			49 CKT B									IN	OUT					5A to 5Y		45	
			50 CKT A												OUT	IN		6A to 6Y		60	
			50 CKT B												OUT	IN		6A to 6Y		45	
10 T_A=125° C	t_{PHL}		51 CKT A	IN													OUT	1A to 1Y		90	
			51 CKT B	IN													OUT	1A to 1Y		60	
			52 CKT A		OUT	IN		IN										2A to 2Y		90	
			52 CKT B		OUT	IN		IN										2A to 2Y		60	
			53 CKT A						OUT				OUT					3A to 3Y		90	
			53 CKT B						OUT				OUT					3A to 3Y		60	
			54 CKT A							IN	OUT							4A to 4Y		90	
			54 CKT B							IN	OUT							4A to 4Y		60	
			55 CKT A									IN	OUT					5A to 5Y		90	
			55 CKT B									IN	OUT					5A to 5Y		60	
			56 CKT A												OUT	IN		6A to 6Y		90	
			56 CKT B												OUT	IN		6A to 6Y		60	
	t_{PLH}		57 CKT A	IN													OUT	1A to 1Y		90	
			57 CKT B	IN													OUT	1A to 1Y		55	
			58 CKT A		OUT	IN		IN										2A to 2Y		90	
			58 CKT B		OUT	IN		IN										2A to 2Y		55	
			59 CKT A						OUT				OUT					3A to 3Y		90	
			59 CKT B						OUT				OUT					3A to 3Y		55	
			60 CKT A							IN	OUT							4A to 4Y		90	
			60 CKT B							IN	OUT							4A to 4Y		55	
			61 CKT A									IN	OUT					5A to 5Y		90	
			61 CKT B									IN	OUT					5A to 5Y		55	
			62 CKT A												OUT	IN		6A to 6Y		90	
			62 CKT B												OUT	IN		6A to 6Y		55	
11	Same tests, terminal conditions and limits as for subgroup 10, except T_A = -55° C.																				

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TABLE III. Group A inspection for device type 06.
Terminal conditions (pins not designated are open).

Subgroup	Symbol	MIL-STD-883 method	Case C Test No.	1 1A	2 1B	3 1Y	14 VCC	6 2Y	4 2A	5 2B	8 3Y	9 3A	10 3B	7 GND	12 4A	13 4B	11 4Y	Meas. terminal	Test limits Min	Max	Unit
1 $T_A = 25°C$	V_{OL}	3007	1	2.0 v	2.0 v	2 mA	4.5 v	2 mA	5.5 v	5.5 v		5.5 v	5.5 v	GND	5.5 v	5.5 v		1Y		0.3	V
			2	5.5 v	5.5 v				2.0 v	2.0 v		5.5 v	5.5 v		2.0 v	2.0 v		2Y			
			3						5.5 v	5.5 v	2 mA	2.0 v	2.0 v					3Y			
			4									5.5 v	5.5 v				2 mA	4Y			
	I_{CEX}		5	0.6 v	4.5 v	5.5 v				4.5 v					5.5 v	5.5 v		1Y		50	μA
			6	4.5 v	0.6 v	5.5 v		5.5 v	0.6 v	0.6 v		4.5 v	4.5 v					1Y			
			7	5.5 v	5.5 v			5.5 v	4.5 v	5.5 v								2Y			
			8								5.5 v	0.6 v	4.5 v					2Y			
			9								5.5 v	4.5 v	0.6 v					3Y			
			10															3Y			
			11												0.6 v	4.5 v	5.5 v	4Y			
			12												4.5 v	0.6 v	5.5 v	4Y			
	I_{IH1}	3010	13	2.4 v	GND		5.5 v		GND	GND		GND	GND		GND	GND		1A		10	
			14	GND	2.4 v				GND	GND		GND						1B			
			15						2.4 v	2.4 v								2A			
			16						GND	2.4 v		2.4 v	2.4 v					2B			
			17									GND						3A			
			18															3B			
			19															4A			
			20															4B			
	I_{IH2}		21	5.5 v	GND				GND	GND		5.5 v	GND		GND	GND		1A		100	
			22	GND	5.5 v				5.5 v	GND		GND						1B			
			23						GND	5.5 v								2A			
			24									5.5 v	GND					2B			
			25									GND						3A			
			26															3B			
			27															4A			
			28															4B			
	I_{IL}	3009	29	0.3 v	0.3 v				5.5 v			5.5 v			5.5 v			1A	-60	-180	
			30	5.5 v	5.5 v				5.5 v									1B			
			31						0.3 v	0.3 v								2A			
			32						5.5 v	5.5 v		0.3 v	0.3 v					2B			
			33									5.5 v						3A			
			34															3B			
			35															4A			
			36															4B			
	I_{CCL}	3005	37															VCC		2.04	mA
	I_{CCH}	3005	38	GND	GND				GND	GND		GND	GND		GND	GND		VCC		0.80	mA
2	Same tests, terminal conditions and limits as for subgroup 1, except $T_A = 125°C$.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except $T_A = -55°C$.																				
9 $T_A = 25°C$	t_{PHL}	3003 (Fig. 6)	39	IN	2.4 v	OUT	5.0 v	OUT	IN	2.4 v	OUT	IN	2.4 v	GND	IN	2.4 v	OUT	1A to 1Y	3	60	ns
			40															2A to 2Y			
			41															3A to 3Y			
			42															4A to 4Y			
	t_{PLH}		43	IN	2.4 v	OUT		OUT		2.4 v	OUT	IN	2.4 v		IN	2.4 v	OUT	1A to 1Y		90	
			44															2A to 2Y			
			45															3A to 3Y			
			46															4A to 4Y			

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case C	1	2	3	14	6	4	5	8	9	10	7	12	13	11	Meas. terminal	Test limits		
			Test No.	1A	1B	1Y	VCC	2Y	2A	2B	3Y	3A	3B	GND	4A	4B	4Y		Min	Max	Unit
10 T_A = 125° C	t_{PHL}	3003 (Fig. 6)	47 48 49 50	IN	2.4 v	OUT	5.0 v	OUT	IN	2.4 v	OUT	IN	2.4 v	GND	IN	2.4 v	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y	3	90	ns
	t_{PLH}		51 52 53 54	IN	2.4 v	OUT		OUT	IN	2.4 v	OUT	IN	2.4 v		IN	2.4 v	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y		140	
11	Same tests, terminal conditions and limits as for subgroup 10, except T_A = -55° C.																				

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5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging and packing. Microcircuits shall be prepared for delivery in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment) and logistic purposes.

6.3 Ordering data. The contract or order should specify the following:

(a) Complete part number (see 1.2).
(b) Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.
(c) Requirements for certificate of compliance, if applicable.
(d) Requirements for notification of change of product or process to procuring activity in addition to notification to the qualifying activity, if applicable.
(e) Requirements for packaging and packing.
(f) Requirements for failure analysis (including required test condition of Method 5003), corrective action and reporting of results, if applicable.
(g) Requirements for product assurance options.
(h) Requirements for carriers, special lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

6.4 Abbreviations, symbols and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-STD-1313, MIL-STD-1331, and as follows:

GND - - - - Electrical ground (common terminal)
V_{IN} - - - - - - Voltage level at an input terminal

6.5 Logistic support. Lead materials and finishes (see 3.5), are interchangeable. Unless otherwise specified, microcircuits procured for Government logistic support will be procured to device class B (see 1.2.2), lead material and finish C (see 3.3), and case outline A (see 1.2.3). There is no substitute for case outline C. Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. Microcircuits covered by this specification unilaterally supersede the following commercial device types:

Device type	Commercial type
01	54L30
02	54L20
03	54L10
04	54L00
05	54L04
06	54L03 (case C only) and 54L01 (case A, B, and D)

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Custodians:
Army - EL
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI
Navy -
Air Force - 11, 70, 80
DSA - ES

User activities:
Army - SM, MU, WC
Navy - AS, OS, CG, MC
Air Force - 13, 15, 19

Preparing activity:
Air Force - 17

Agent:
DSA - ES

(Project 5962-T041)

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